

A 4.0-N puck is traveling at 3.0m/s. It strikes a 8.0-N puck, which is stationary. The two pucks stick together. Their common final speed is:

- ▶ 1.0m/s
- ▶ 1.5m/s
- ▶ 2.0m/s
- ▶ 2.3m/s

An object moving in a circle at constant speed:

- ▶ must have only one force acting on it
- ▶ is not accelerating
- ▶ is held to its path by centrifugal force
- ▶ has an acceleration of constant magnitude

A plane traveling north at 200m/s turns and then travels south at 200m/s.

The change in its velocity is:

- ▶ 400m/s north
- ▶ 400m/s south
- ▶ zero
- ▶ 200m/s south

At time $t = 0$ a car has a velocity of 16 m/s. It slows down with an acceleration given by $-0.50t$, in m/s^2 for t in seconds. It stops at $t =$

- ▶ 64 s
- ▶ 32 s
- ▶ 16 s

▶ 8.0 s

1 mi is equivalent to 1609 m so 55 mph is:

▶ 15 m/s

▶ 25 m/s

▶ 66 m/s

▶ 88 m/s

The number of significant figures in 0.00150 is:

▶ 5

▶ 4

▶ 3

▶ 2

One revolution is the same as:

▶ 1 rad

▶ 57 rad

▶ $\pi/2$ rad

▶ π rad

▶ 2π rad

For a body to be in equilibrium under the combined action of several forces:

▶ All the forces must be applied at the same point

▶ all of the forces form pairs of equal and opposite forces

▶ any two of these forces must be balanced by a third force

▶ the sum of the torques about any point must equal zero

A bucket of water is pushed from left to right with increasing speed across a horizontal surface. Consider the pressure at two points at the same level in the water.

- ▶ It is the same
- ▶ It is higher at the point on the left
- ▶ It is higher at the point on the right
- ▶ At first it is higher at the point on the left but as the bucket speeds up it is lower there

An organ pipe with both ends open is 0.85m long. Assuming that the speed of sound is 340m/s, the frequency of the third harmonic of this pipe is:

- ▶ A. 200 Hz
- ▶ B. 300 Hz
- ▶ C. 400 Hz
- ▶ D. 600 Hz

Capacitors C1 and C2 are connected in series. The equivalent capacitance is given by

- ▶ $C_1 C_2 / (C_1 + C_2)$
- ▶ $(C_1 + C_2) / C_1 C_2$
- ▶ $1 / (C_1 + C_2)$
- ▶ C_1 / C_2

If the potential difference across a resistor is doubled:

- ▶ only the current is doubled

- ▶ only the current is halved
- ▶ only the resistance is doubled
- ▶ only the resistance is halved

By using only two resistors, R_1 and R_2 , a student is able to obtain resistances of 3Ω , 4Ω , 12Ω , and 16Ω . The values of R_1 and R_2 (in ohms) are:

- ▶ 3, 4
- ▶ 2, 12
- ▶ 3, 16
- ▶ 4, 12

Faraday's law states that an induced emf is proportional to:

- ▶ the rate of change of the electric field
- ▶ the rate of change of the magnetic flux
- ▶ the rate of change of the electric flux
- ▶ the rate of change of the magnetic field

A generator supplies 100V to the primary coil of a transformer. The primary has 50 turns and the secondary has 500 turns. The secondary voltage is:

- ▶ 1000V
- ▶ 500V
- ▶ 250V
- ▶ 100V

Which of the following electromagnetic radiations has photons with the greatest energy?

- ▶ blue light
- ▶ yellow light
- ▶ x rays
- ▶ radio waves

A virtual image is one:

- ▶ toward which light rays converge but do not pass through
- ▶ from which light rays diverge as they pass through
- ▶ toward which light rays converge and pass through
- ▶ from which light rays diverge but do not pass through

What is the unit of magnification factor?

- ▶ meter.Kelvin
- ▶ radian.Kelvin
- ▶ degree.Kelvin
- ▶ no units

During an adiabatic process an object does 100 J of work and its temperature decreases by 5K. During another process it does 25 J of work and its temperature decreases by 5 K. Its heat capacity for the second process is.

- ▶ 20 J/K
- ▶ 100 J/K
- ▶ 15 J/K
- ▶ 5 J/K

An ideal gas expands into a vacuum in a rigid vessel. As a result there is:

- ▶ a change in entropy
- ▶ a decrease of internal energy
- ▶ an increase of pressure
- ▶ a change in temperature

The Stern-Gerlach experiment makes use of:

- ▶ a strong uniform magnetic field
- ▶ a strong non-uniform magnetic field
- ▶ a strong uniform electric field
- ▶ a strong non-uniform electric field

A large collection of nuclei are undergoing alpha decay. The rate of decay at any instant is proportional to:

- ▶ the number of undecayed nuclei present at that instant
- ▶ the time since the decays started
- ▶ the time remaining before all have decayed
- ▶ the half-life of the decay

As a 2.0-kg block travels around a 0.50-m radius circle it has an angular speed of 12 rad/s. The circle is parallel to the xy plane and is centered on the z axis, a distance of 0.75m from the origin. The z component of the angular momentum around the origin is:

- ▶ $6.0 \text{ kg} \cdot \text{m}^2/\text{s}$
- ▶ $9.0 \text{ kg} \cdot \text{m}^2/\text{s}$
- ▶ $11 \text{ kg} \cdot \text{m}^2/\text{s}$

► $14 \text{ kg} \cdot \text{m}^2/\text{s}$

Questions:

Question No: 1 (Marks: 3)

Two people are carrying a uniform wooden board that is 3.00 m long and weighs 160 N. If one person applies an upward force equal to 60 N at one end, at what point does the other person lift? Begin with a free-body diagram of the board.

Forces in x direction = 0

Forces in Y = $F_1 + F_2 - W$

Question No: 2 (Marks: 3)

If a charged particle moves in a straight line through some region of space, can you say that the magnetic field in that region is zero?

Question No: 3 (Marks: 5)

A bike accelerates uniformly from rest to a speed of 7.10 m/s over a distance of 35.4 m. Determine the acceleration of the bike.

Question No: 4 (Marks: 5)

A flat loop of wire consisting of a single turn of cross-sectional area 8.00 cm^2 is perpendicular to a magnetic field that increases uniformly in magnitude from

0.500 T to 2.50 T in 1.00 s. What is the resulting induced current if the loop has a resistance of 2.00 Ω ?

VU Insider